

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO3: Analyse knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I R. Rupa Sree (192472122) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. Rupa Sree

25+25+18+18+9
E 95
AP

Question 1

Case Study - Cybersecurity Intrusion Detection System

A company is designing a rule-based Logical Agent to detect network intrusions. The system's knowledge base contains:

- I. If there is FailedLoginAttempts AND UnusualIPAddress $\rightarrow$ Possible BruteForceAttack
- II. If there is FailedLoginAttempts AND OddLoginTime $\rightarrow$ Possible AccountCompromise
- III. If there is UnusualIPAddress AND HighDataTransfer $\rightarrow$ Possible DataExfiltration
- IV. Facts about Session X: FailedLoginAttempts = True, UnusualIPAddress = True, HighDataTransfer = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible security alerts for Session X. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Session X indicates DataExfiltration.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Question 2

Case Study - Airport Priority Landing System

An airport control tower uses a FOL-based knowledge base to manage priority landing clearance:

- I. for-all x: Aircraft(x) AND EmergencyDeclared(x) $\rightarrow$ PriorityClearance(x) [Rule 1]
- II. for-all x: PriorityClearance(x) $\rightarrow$ RunwayReserved(x) AND NOT HoldPattern(x) [Rule 2]
- III. for-all x for-all y: Aircraft(x) AND QueuedAfter(x,y) AND RunwayReserved(y) $\rightarrow$ Delay(x) [Rule 3]
- IV. Facts: Aircraft(FlightZ), EmergencyDeclared(FlightZ), Aircraft(FlightY), QueuedAfter(FlightY, FlightZ)

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set theta (MGU) at each step.
- (b) Using the Resolution method, prove that 'Delay(FlightY)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency aircraft. Identify what additional rules or facts are needed and justify logically.

Question 3

Case Study - Industrial Equipment Maintenance Expert System

A factory maintenance logical agent operates with the following propositional knowledge base (KB):

- 1) P1: MachineOverheating $\rightarrow$ MaintenanceNeeded
- 2) P2: MaintenanceNeeded AND HighPriorityLine $\rightarrow$ ScheduleImmediateRepair
- 3) P3: NOT ScheduleImmediateRepair AND HighPriorityLine $\rightarrow$ ProductionRisk
- 4) Facts: MachineOverheating = True, HighPriorityLine = True

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move NOT inward $\rightarrow$ distribute).

- (b) Using Resolution Refutation, prove the theorem 'ScheduleImmediateRepair'. Negate the goal, combine with the KB in CNF, and derive the empty clause. Draw the resolution proof tree.
- (c) If the fact 'MachineOverheating' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'ProductionRisk' still follow? Apply appropriate reasoning patterns to justify.

Question 4

Case Study - Minefield Robot Knowledge Agent

An AI robot navigating a minefield-clearance grid perceives: Vibration at [2,1], Smoke at [1,1], Beacon at [2,2]. The agent uses a propositional KB with these rules:

- 1) Vibration[2,1] $\rightarrow$ Mine is adjacent to cell [2,1]
- 2) Smoke[1,1] $\rightarrow$ Fire is adjacent to cell [1,1]
- 3) Beacon[x,y] $\rightarrow$ SafeZoneMarker is at [x,y]
- 4) NOT Mine[x,y] AND NOT Fire[x,y] $\rightarrow$ Safe[x,y]

- (a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.
- (b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Mine and Fire. Show each reasoning step precisely.
- (c) Evaluate whether the path [1,1] $\rightarrow$ [2,1] $\rightarrow$ [2,2] is safe for the robot to reach the SafeZoneMarker. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks (100)
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly	25
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	25
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	18
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	18
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	9

95

95

1) Cybersecurity Intrusion Detection System:-

a) Representing using propositional logic:-

F = Failed login Attempts

U = Unusual Address

O = Odd login time

H = High data transfer

B = Possible Brute force Attack

C = Possible account compromise

D = Possible data exfiltration.

Rules:-

$$1) F \wedge U \rightarrow B \rightarrow F \rightarrow \text{True}$$

$$2) F \wedge O \rightarrow C \rightarrow U \rightarrow \text{True}$$

$$3) U \wedge H \rightarrow D \rightarrow H \rightarrow \text{True}$$

b) Forward chaining:-

Step-1 $\rightarrow$ known - F, U, H

$$F \wedge U \rightarrow B$$

possible Brute force attack

Step-2 $\rightarrow F \wedge O \rightarrow C$ $O \rightarrow$ is not given

This cannot be fire

Step-3:- $U \wedge H \rightarrow D$

possible Data exfiltration

c) Backward chaining:-

Goal:- possible Data exfiltration

$$U \wedge H \rightarrow D$$

$$U \wedge H \rightarrow \text{True}$$

$$D \leftarrow U \wedge H$$

Both subgoals are satisfied.

2) Airport Priority Landing System :-

@ Unification

$\forall x (\text{Aircraft}(x) \wedge \text{EmergencyDeclared}(x)) \rightarrow$
 $\rightarrow \text{priority clearance}(x)$

convert to :

$\text{Aircraft}(x) \wedge \text{EmergencyDeclared}(x) \rightarrow$
 $\text{priority clearance}(x)$

Given Unification is

$$\theta = \{x / \text{Flight } z\}$$

Therefore

$\text{priority clearance}(\text{Flight } z)$

b) Resolution Proof :-

CNF Rules :-

Rule 1 :-

$\neg \text{Aircraft}(x) \vee \neg \text{EmergencyDeclared}(x) \vee$
 $\text{priority clearance}(x)$

$\therefore \text{priority clearance}(\text{Flight } z)$

Rule 2 :- $\neg \text{priority clearance}(x) \vee \text{RunwayReserved}(x)$

$\therefore \text{RunwayReserved}(\text{Flight } z)$

Rule 3 :-

$\neg \text{Aircraft}(x) \vee \neg \text{DelayedAfter}(x, y) \vee \neg \text{Runway}$
 $\text{Reserved}(y) \vee \text{Delay}(x)$

$\therefore \text{Delay}(\text{Flight } y)$

For resolution refutation : $\neg \text{Delay}(\text{Flight } y)$

Combined with derived : $\text{Delay}(\text{Flight } y)$

So it is proven

c) Simultaneous Emergency Aircraft :-
Current KB can identify emergency aircraft,
but it does not affect runway conflicts
between two emergency aircraft.

Additional rules are needed for:

- > Runway availability
- > Runway assignment
- > Conflict resolution

Thus, the KB should include rules to simulate
emergency flights.

3. Industrial Equipment Maintenance Expert System.

a) CNF Conversion

M = Machine Overheating

N = Maintenance Needed

H = High Priority line

S = Schedule Immediate Repair

R = production Risk

P1 :- $M \rightarrow N$ $\neg M \vee N$

P2 :- $N \wedge H \rightarrow S$ $\neg N \vee \neg H \vee S$

P3 :- $\neg S \wedge H \rightarrow R$ $S \vee \neg H \vee \neg R$

Facts :- M, H

b) Resolution Refutation

Goal : S

Negate Goal : $\neg S$

From : M, $\neg M \vee N$

we get : N

Then:- $N, \neg N \vee H \vee S$

Gives:- $\neg H \vee S$

using H: S

Finally $S + \neg S \rightarrow 0$

$\therefore$ Schedule Immediate Repair is proven

c) Remove Machine Overheating:-

If M is removed:

- > Maintenance Needed cannot be derived

- > Schedule Immediate Repair cannot be derived

$\therefore$ production Risk does not follow.

4) Minefield Robot Knowledge Agent:-

a) Belief State

Initial percepts:

Vibration[2,1], Smoke[1,1], Beacon[2,2]

using Modus Ponens:

Vibration[2,1] $\rightarrow$ Mine Adjacent[2,1]

Smoke[1,1] $\rightarrow$ Fire Adjacent[1,1]

Beacon[2,2] $\rightarrow$ Safezone Marker[2,2]

Therefore derived facts are:

Mine Adjacent[2,1]

Fire Adjacent[1,1]

Safezone Marker[2,2]

b) Forward Chaining :-

1) Vibration $[2,1] \rightarrow$ Mine Adjacent $[2,1]$

2) Smoke $[1,1] \rightarrow$ Fire Adjacent $[1,1]$

3) Beacon $[2,2] \rightarrow$ Safe zone Marker $[2,2]$

So,

•) Mine : Adjacent to $[2,1]$

•) Fire : Adjacent to $[1,1]$

•) Safe zone : $[2,2]$

The exact mine and fire locations cannot be determined from given information.

c) path Safety :-

path: $[1,1]$

↓

$[2,1]$

↓

$[2,2]$

•) $[1,1]$ has nearby fire.

•) $[2,1]$ has nearby mine

•) $[2,2]$ is Safe zone Marker

Therefore, the complete path cannot be confirmed as safe